



CLASS ASSIGNMENT #02

AI

(Artificial Intelligence)

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Question 1: Applications of Genetic Algorithms

Genetic Algorithms (GAs) are powerful problem-solving techniques inspired by the process of natural selection. They are especially useful for tackling complex optimization problems where the number of possible solutions is too vast to check one by one. GAs work by evolving a population of candidate solutions over many generations to find the best possible outcome.

Below are three examples of problems from different fields that can be solved effectively using Genetic Algorithms.

Example 1: The Traveling Salesperson Problem (Logistics)

The Traveling Salesperson Problem (TSP) is a most common challenge in logistics and computer science where the goal is to find the most efficient route possible.

Component	Description
Problem	To find the shortest possible route that visits a list of cities exactly once and returns to the starting city.
Initial Population	A collection of randomly generated routes. Each route is a complete and valid tour that visits every city.
Representation	A single solution (a "chromosome") is represented as an ordered list of cities, for example: [City D, City A, City C, City B].
Evaluation (Fitness)	The fitness of a route is determined by its total distance. To find the shortest path, we aim to maximize a fitness score calculated as 1 / Total Distance . A shorter distance results in a higher fitness.
Crossover	Two "parent" routes are combined to create a "child" route. A common method is Order Crossover (OX) , where a section of one parent's route is copied to the child, and the remaining cities are filled in from the second parent in the order they appear. This ensures every city is visited exactly once.
Mutation	A small, random change is introduced into a route to explore new possibilities. For example, Swap Mutation involves randomly picking two cities in a single route and swapping their positions.
Exit Criteria	The algorithm stops when a certain condition is met, such as after running for a fixed number of generations (e.g., 500), or when the best route found has not improved for a set number of generations.

Example 2: University Timetabling (Education & Administration)

Creating a university timetable is a complex scheduling problem with many conflicting rules and preferences that GAs are well-suited to handle.

Component	Description
Problem	To assign every course to a specific time slot, classroom, and professor, while minimizing conflicts. Conflicts include "hard" rules such as two classes cannot be in the same room at the same time and "soft" preferences such as, avoiding 8 AM classes.
Initial Population	A collection of randomly generated, complete timetables. These initial timetables might contain many conflicts.
Representation	A single solution is a list containing an assignment for every course, such as: [(CS101, Prof. Ahmed, Room C-301, Mon 9 AM), (EE204, ...)].
Evaluation (Fitness)	The fitness is calculated based on a penalty system. Hard conflicts are given a very high penalty, while soft conflicts receive a lower penalty. The goal is to maximize fitness by minimizing the total penalty score, ideally reaching a penalty of zero.
Crossover	Sections of two parent timetables are swapped to create new child timetables. For instance, all Monday morning assignments from Parent 1 could be combined with the remaining assignments from Parent 2.
Mutation	A random change is made to a single class assignment in a timetable, such as moving it to a different time slot or classroom.
Exit Criteria	The algorithm stops when a timetable with a penalty score of zero (meaning no hard conflicts) is found, or after a certain amount of time or number of generations has passed.

Example 3: Financial Portfolio Optimization (Finance)

In finance, GAs can help investors build a portfolio of assets (like stocks and bonds) that balances risk and reward according to their goals.

Component	Description
Problem	To decide how much capital to invest in a set of different assets to maximize the expected financial return for a given level of risk.
Initial Population	A collection of randomly generated portfolios. Each portfolio is a list of weights representing the percentage of investment in each asset, with the weights always summing to 100%.
Representation	A single solution is a list of numerical weights, for example: [0.40, 0.25, 0.35], representing 40% in Asset A, 25% in Asset B, and 35% in Asset C.
Evaluation (Fitness)	The fitness of a portfolio is its risk-adjusted return. A common formula is Expected Return – (Risk Tolerance × Risk Level) . A higher score indicates a better portfolio.

Crossover	A new child portfolio is created by taking a weighted average of two parent portfolios. For example, $\text{Child} = (0.7 * \text{Parent 1}) + (0.3 * \text{Parent 2})$. The new weights are then adjusted to ensure they sum to 100%.
Mutation	One or more weights in a portfolio are slightly changed. For instance, a small amount might be moved from one asset to another.
Exit Criteria	The algorithm typically stops after a fixed number of generations, or when the fitness of the best portfolio in the population no longer shows significant improvement.

Question 2: The 8 Queens Problem

The goal of the 8 Queens problem is to place eight chess queens on an 8x8 chessboard so that no two queens can attack each other. This means no two queens can share the same row, column, or diagonal. This puzzle can be modeled and solved as a Constraint Satisfaction Problem (CSP).

1. Problem Formulation as a CSP

A CSP is defined by three components: variables, domains, and constraints.

- **Variables:** We can define one variable for each column, Q_1 through Q_8 . The variable Q_i represents the queen in column i .
- **Domains:** The domain for each variable is the set of possible rows where a queen can be placed, which is $\{1, 2, 3, 4, 5, 6, 7, 8\}$. An assignment like $Q_2 = 3$ means the queen in the second column is placed in the third row.
- **Constraints:** For any two queens, Q_i and Q_j :
 1. **Row Constraint:** They cannot be in the same row ($Q_i \neq Q_j$).
 2. **Diagonal Constraint:** They cannot be on the same diagonal ($|i - j| \neq |Q_i - Q_j|$).

2. Solution Strategy: Backtracking Search

This problem can be solved efficiently using a **backtracking search** algorithm. The strategy is as follows:

1. **Start:** Begin at the first column (Q_1) and place a queen in the first valid row (row 1).
2. **Proceed:** Move to the next column (Q_2) and try to place a queen in a row where it does not conflict with any previously placed queens.
3. **Check:** Continue this process column by column. For each new column, find a valid row.

4. **Backtrack:** If you reach a column where no row is valid (because every row is under attack by another queen), you have reached a dead end. In this case, you "backtrack" to the previous column and move its queen to the *next* valid row.
5. **Finish:** The search is complete when a valid row has been found for the queen in the final column (Q8).

One possible solution is the assignment [4, 2, 7, 3, 6, 8, 5, 1], which corresponds to:

- Q1 in row 4
- Q2 in row 2
- Q3 in row 7
- Q4 in row 3
- Q5 in row 6
- Q6 in row 8
- Q7 in row 5
- Q8 in row 1

Question 3: B.S. Data Science Scheduling Problem

Q3.1: Modeling the Problem as a CSP

The problem of finding a valid set of courses for Mateen to graduate can be modeled as a CSP.

- **Variables:** The four graduation requirements are the variables.
 - V_Alg: Algorithms Requirement
 - V_ML: Machine Learning Requirement
 - V_Com: Communications Requirement
 - V_ICT: ICT Requirement
- **Domains:** The initial domains for each variable are the sets of courses that can satisfy it.
 - D_Alg = {{10, 11}, {10, 12}, {13, 15}}
 - D_ML = {{12}, {13}, {15}}
 - D_Com = {{15}, {17}, {16}}
 - D_ICT = {{12}, {18}}

- **Constraints:**

1. **Course Uniqueness:** A course cannot be used to satisfy more than one requirement.
2. **Restriction {12, 13, 18}:** A student can take at most one course from this set.
3. **Arithmetic Restriction {10, 16}:** A student cannot take both course 10 and course 16.
4. **Organization Restriction {15, 17}:** A student cannot take both course 15 and course 17.

Q3.2: Solving with Constraint Propagation

We are given that Mateen has already taken course **18** for his ICT requirement and course **10** towards his Algorithms requirement. We can use this information to reduce the possible options.

Step 1: Apply Initial Information

- Mateen took **18** for V_ICT. This forces the assignment $V_ICT = \{18\}$.
- According to **Constraint 2**, since 18 is taken, courses 12 and 13 are now forbidden.

Step 2: Prune Domains Based on Forbidden Courses

We now remove any options containing 12 or 13 from all other domains.

- D_Alge was $\{\{10, 11\}, \{10, 12\}, \{13, 15\}\}$. After removing options with 12 or 13, and knowing 10 must be included, the only option left is $\{10, 11\}$.
 - **New D_Alge = $\{\{10, 11\}\}$**
- D_ML was $\{\{12\}, \{13\}, \{15\}\}$. After removing 12 and 13, the only option left is $\{15\}$.
 - **New D_ML = $\{\{15\}\}$**

Step 3: Make Forced Assignments

The domains for V_Alge and V_ML now have only one possible choice, forcing the following assignments:

- $V_Alge = \{10, 11\}$
- $V_ML = \{15\}$

Step 4: Propagate Final Constraints on V_Com

So far, Mateen's required courses are $\{10, 11, 15, 18\}$. Let's check the remaining domain for V_Com, which is $\{\{15\}, \{17\}, \{16\}\}$.

- The option $\{15\}$ is now **invalid** because course 15 is already being used for the V_ML requirement (**Constraint 1**).

- The option {17} is now **invalid** because Mateen is taking course 15, and students cannot take both 15 and 17 (**Constraint 4**).
- The option {16} is now **invalid** because Mateen is taking course 10, and students cannot take both 10 and 16 (**Constraint 3**).

After applying all constraints, every possible option for the Communications requirement (V_Com) has been eliminated. The domain D_Com is now empty.

Conclusion

No, a valid schedule cannot be created. The process of constraint propagation shows that Mateen's initial choices of taking course **18** for ICT and course **10** for Algorithms make it impossible to find a set of courses that satisfies all remaining graduation requirements.